

Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

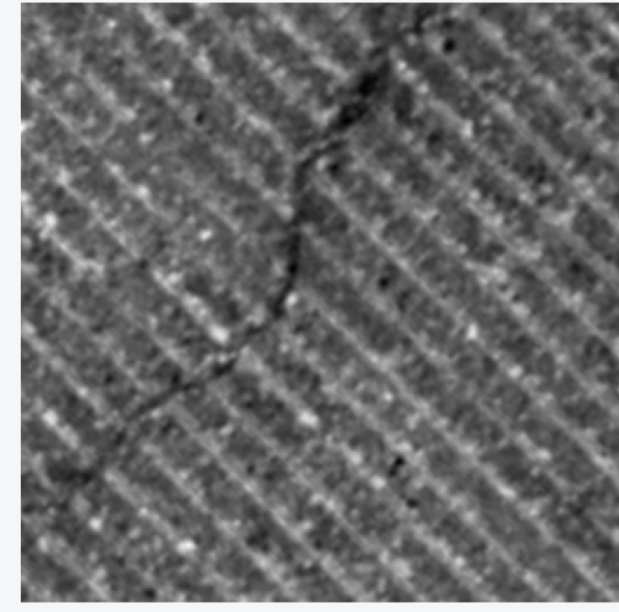
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¹Inria, Université Côte d'Azur, Ayana team, France

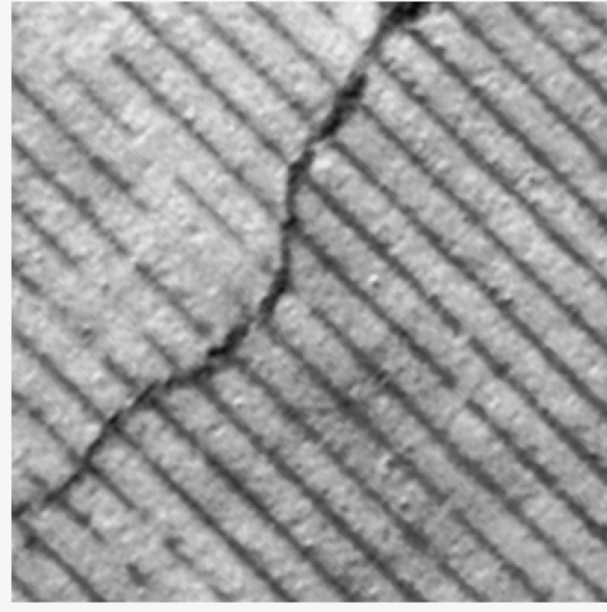
²Cerema, ENDSUM research team, France

From aligned images to a crack skeleton: without training

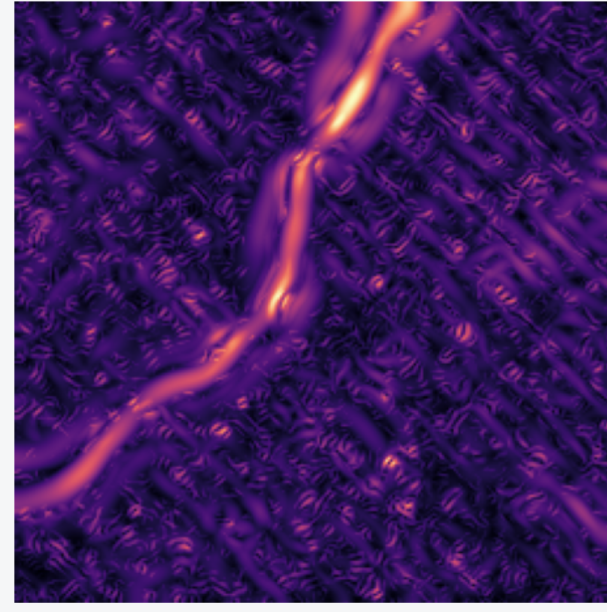
Idea: combine the geometry seen by several sensors, link compatible pixels, then extract the main centerline. No annotated training set is required.



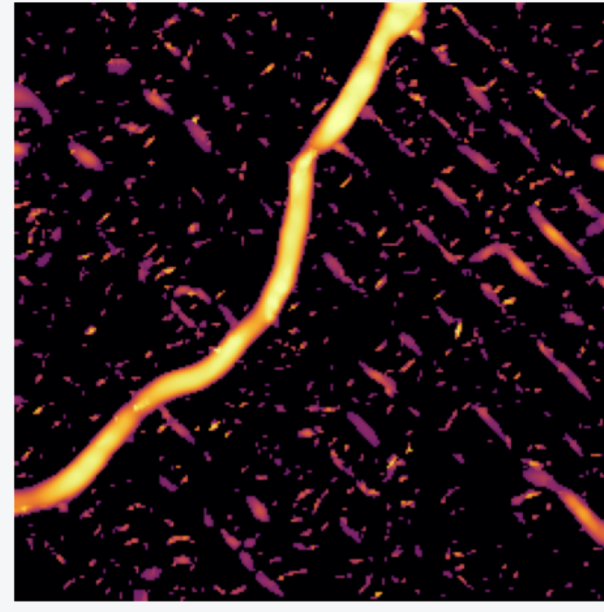
Intensity I



Range R

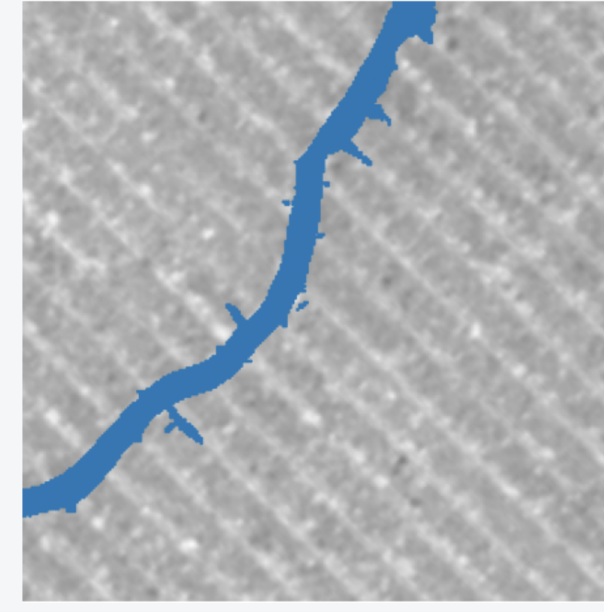


Positive curvature

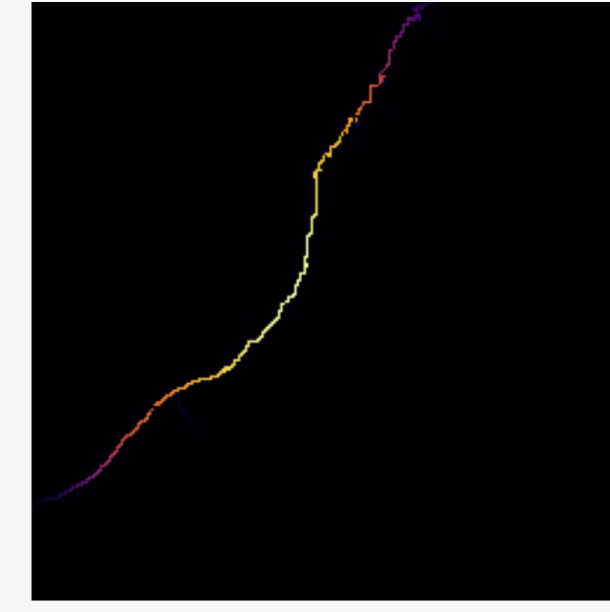


Graph similarity

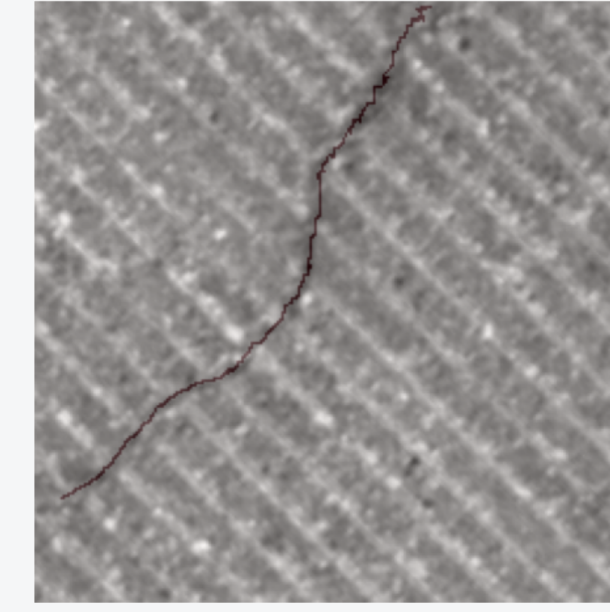
FIND example from the paper. Similarity: maximum over incident edges. Red: extracted skeleton.



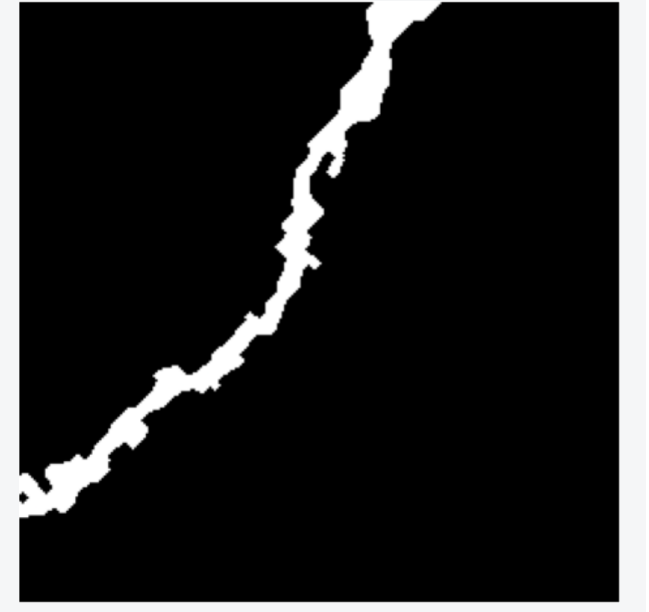
Largest component



Tree centrality

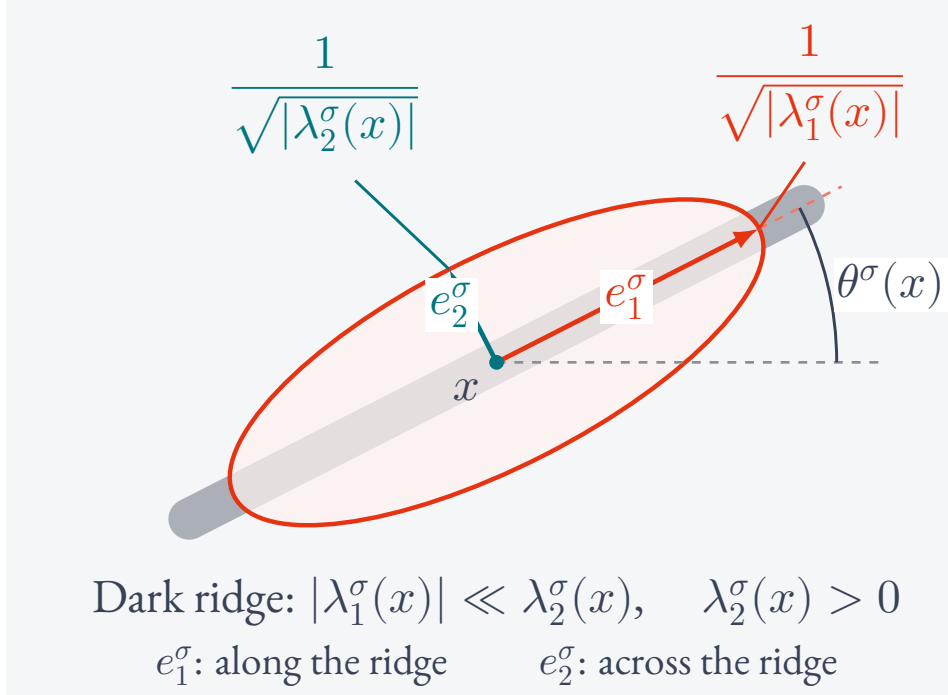


Our skeleton $\hat{\Gamma}$



Ground truth

1. Fuse the geometry across modalities

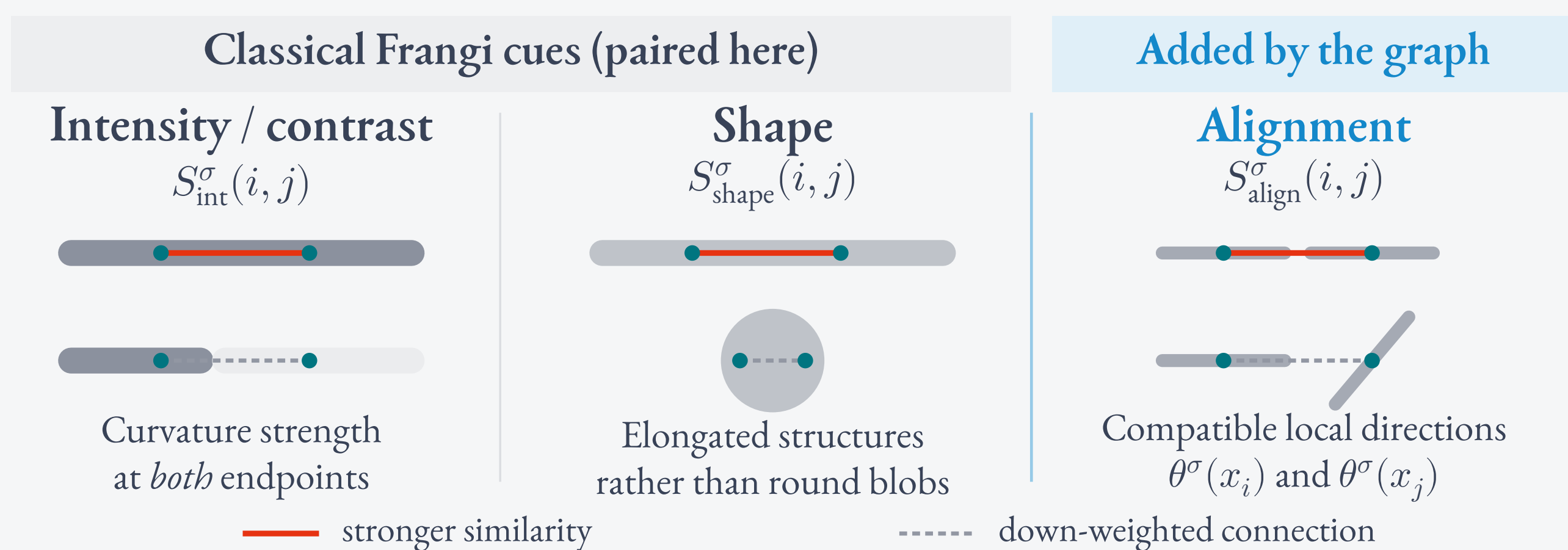


The Hessian $\mathcal{H}_\sigma^{(m)}$ describes ridge geometry at scale σ . $|\lambda_1^\sigma| \leq |\lambda_2^\sigma|$: the first eigenvector follows the ridge; θ^σ gives its direction.

$$\hat{\mathcal{H}}_\sigma^{(m)}(x) = \frac{\mathcal{H}_\sigma^{(m)}(x)}{\max_{y \in \Omega} \|\mathcal{H}_\sigma^{(m)}(y)\|_2}, \quad \mathcal{H}_\sigma^{\text{fused}} = \sum_{m \in \mathcal{M}} w_m \hat{\mathcal{H}}_\sigma^{(m)}.$$

$w_m \geq 0, \sum_m w_m = 1$. Recompute the eigensystem after fusion. Inputs must be aligned and share a dark-ridge polarity.

2. From pixel-wise Frangi [1] to pairwise links



Contrast and shape reuse classical Frangi cues. **The graph adds alignment.** For nearby pixels, $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

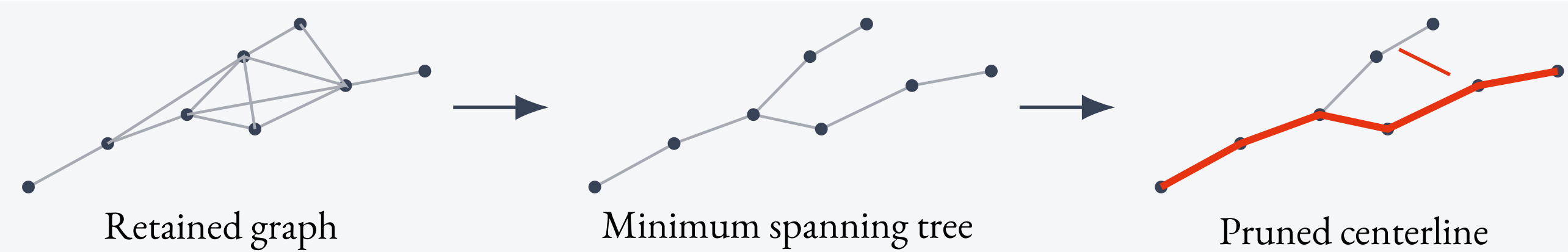
$$C_\sigma(x) = \max\{\lambda_2^\sigma(x), 0\}, \quad R_B^\sigma(x) = \frac{|\lambda_1^\sigma(x)|}{|\lambda_2^\sigma(x)|}.$$

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^\sigma(i, j) S_{\text{shape}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j)$$

$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}.$$

ρ_{ij} penalizes longer links. Directions are unoriented: θ and $\theta + \pi$ describe the same axis.

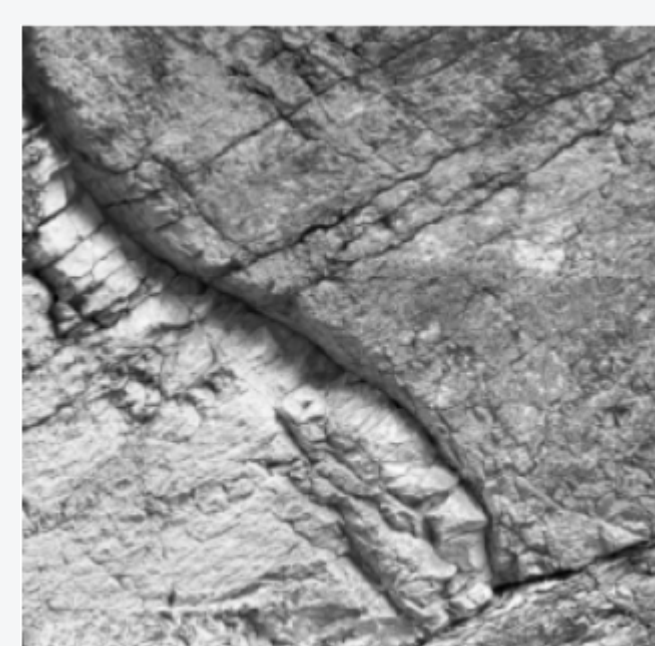
3. Reduce the graph to its main centerline



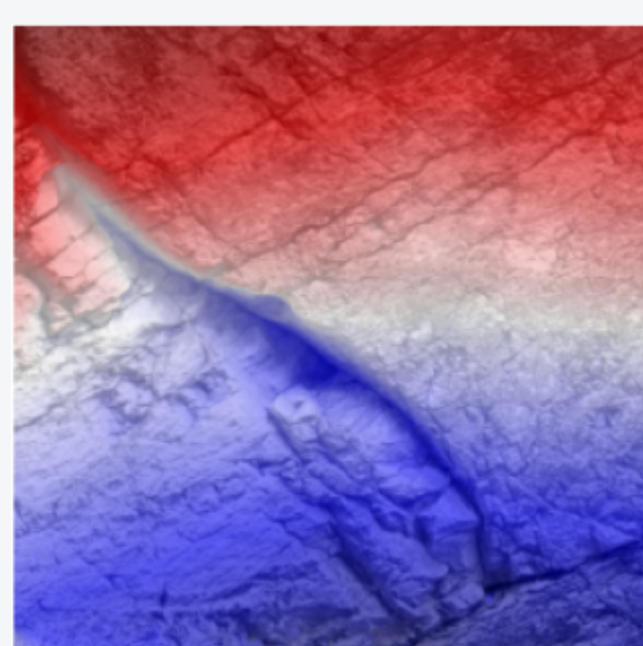
Retain strong edges and nodes using S_{ij} ; select components. Build one tree $\mathcal{T}$ per component using costs d_{ij} .

Similarity-weighted tree centrality selects the main axis and prunes peripheral branches. Diagram is schematic. Settings are selected per dataset; on FIND, $\Sigma = \{1, 3, 5, 7\}$, $R = 3$, retained fractions $\tau_E = \tau_V = 0.25$.

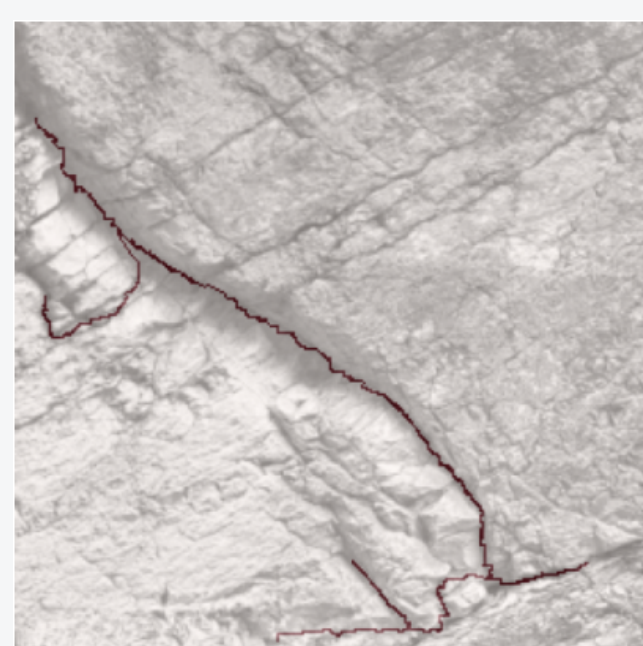
A second setting: a fractured limestone wall



Intensity



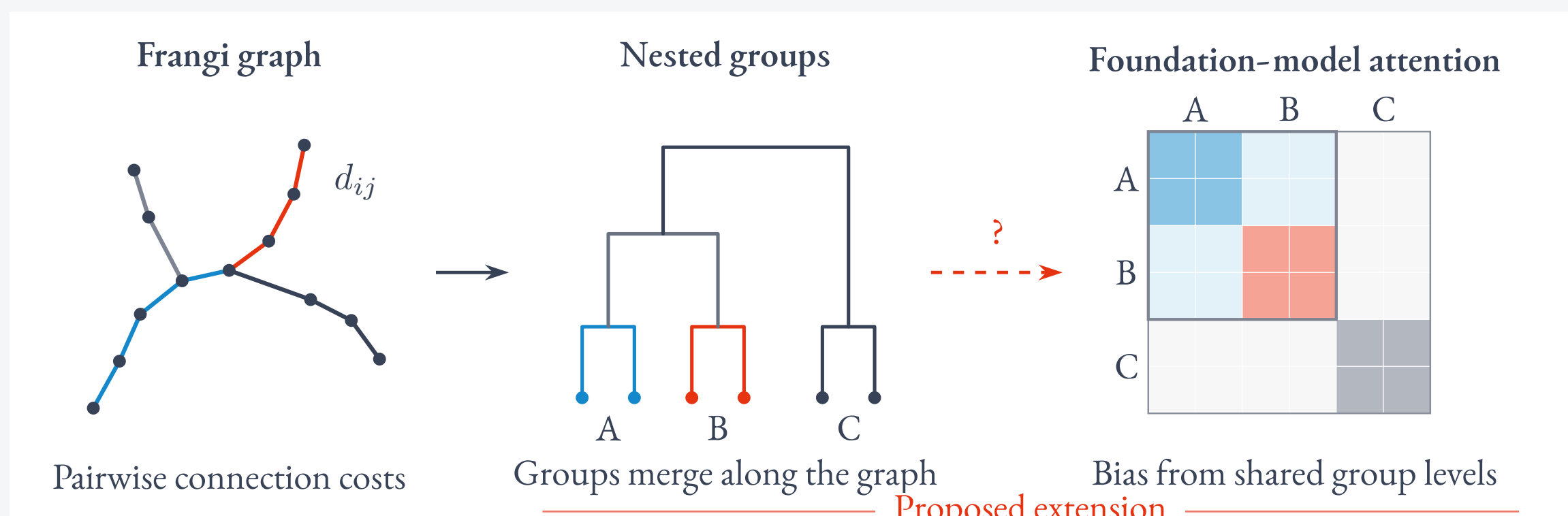
Depth overlay



Our skeleton

Palais des Papes. Main fracture recovered without retraining; minor cracks missed at low contrast or depth resolution.

Outlook: could a Frangi hierarchy guide a foundation model?



4. FIND: clean accuracy and noise sensitivity

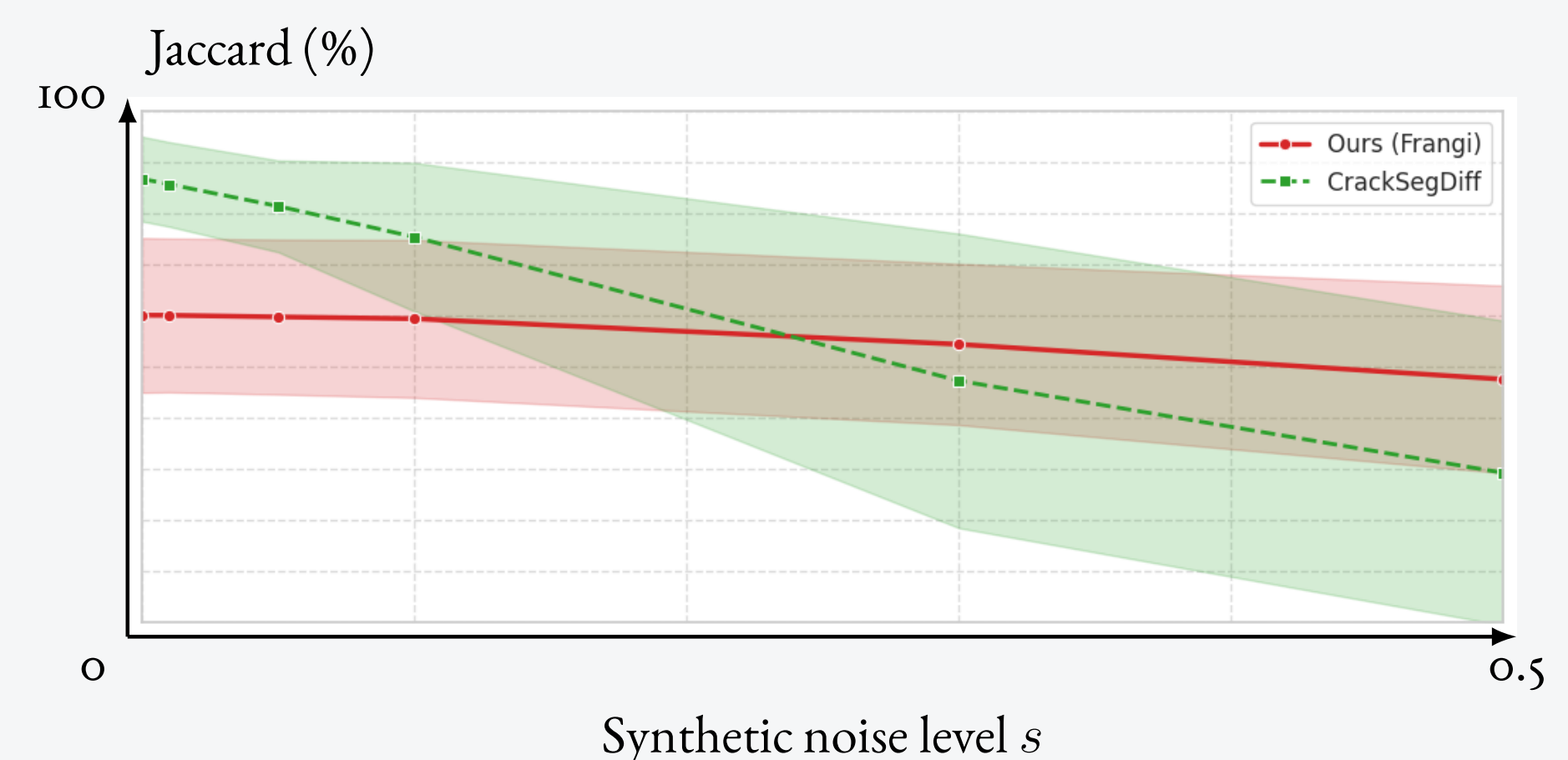
Clean images: the supervised model performs best.

Method ($I + R$)	Jaccard $\uparrow$	Tversky $\uparrow$	$W \downarrow$
Ours, no training	60%	68%	18 px
CrackSegDiff [3]	87%	90%	5 px

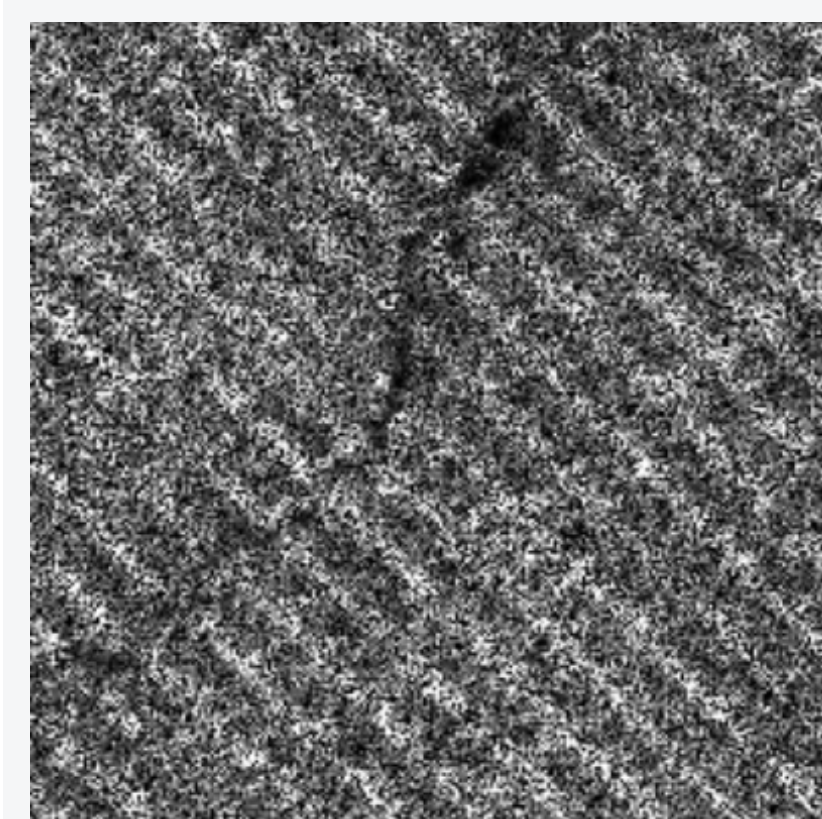
FIND [2], 256×256 . Both skeletons are dilated by 3 px for overlap scores; Tversky: $\alpha = 1, \beta = 0.5$. W : Wasserstein distance between skeleton pixels.

Fusion: intensity alone gives 41% Jaccard; $I + R$ gives 60%. Adding filtered range F gives 63% (equal weights). F is a post-processing of R .

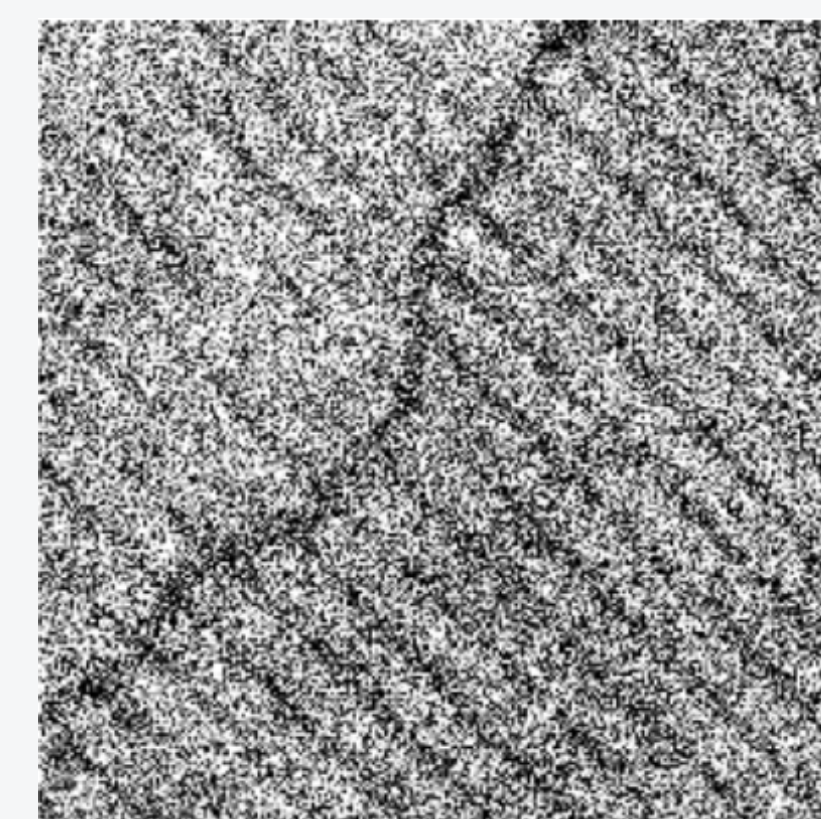
Under the tested noise, our scores degrade more slowly.



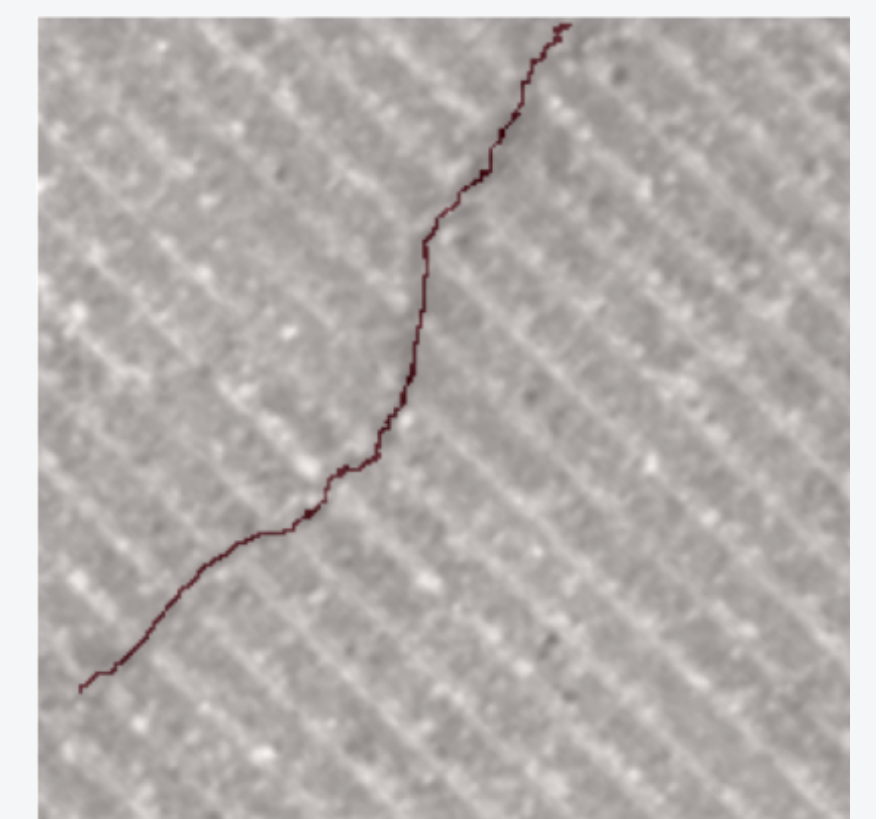
Bands: standard deviation. Curves from the paper.



Noisy intensity



Noisy range

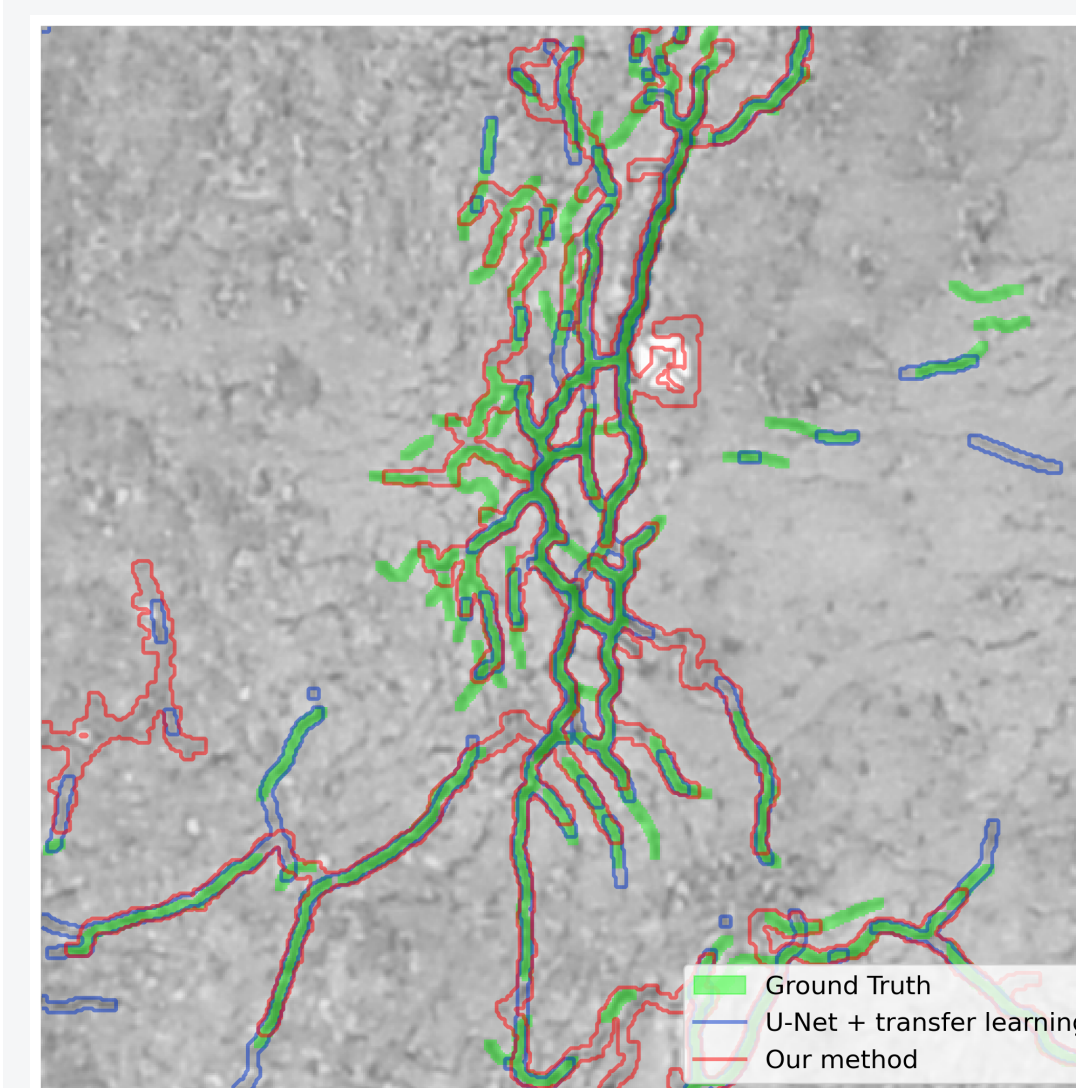


Our skeleton

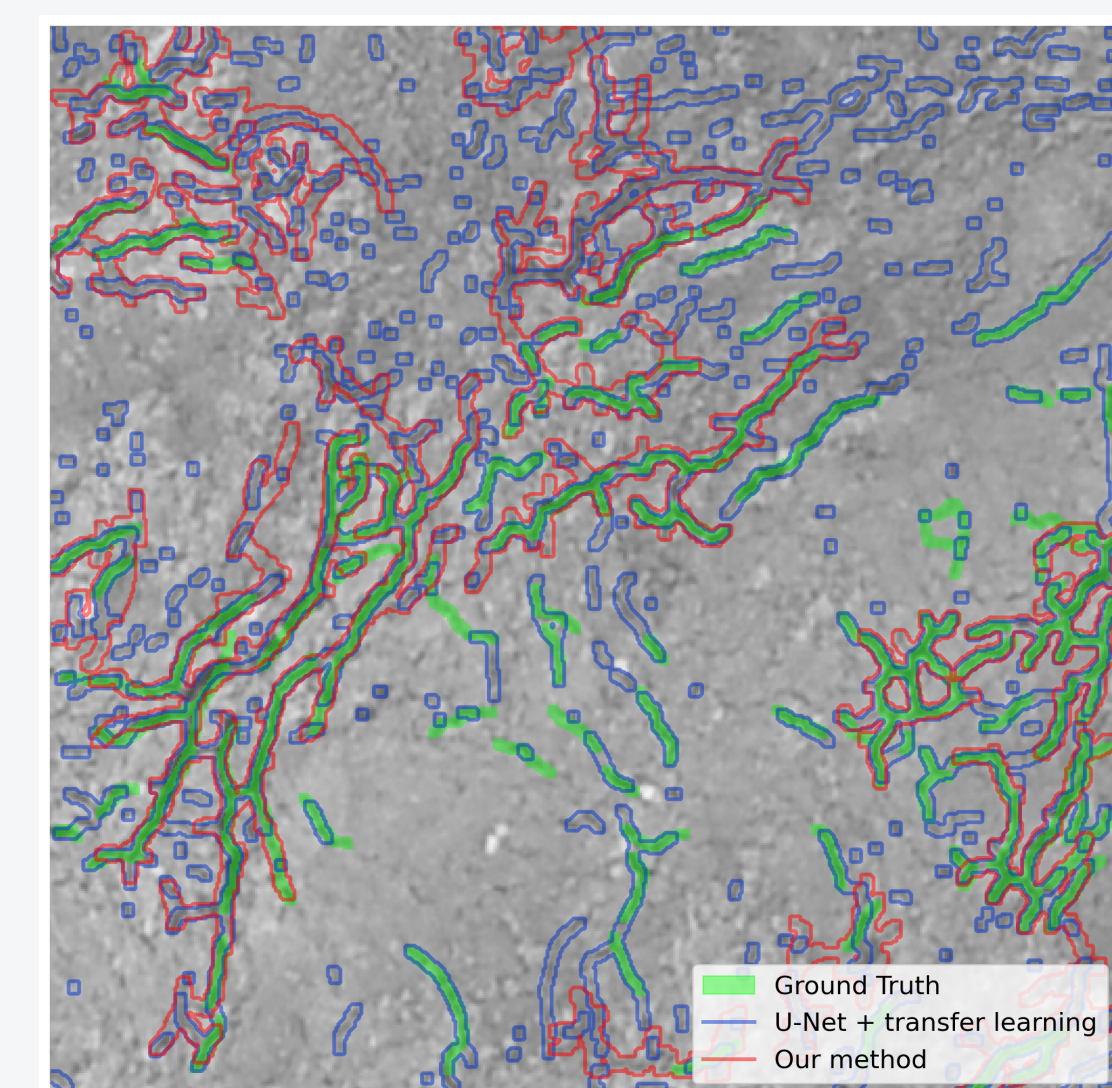
Example at $s = 0.5$: multiplicative speckle on intensity, additive Gaussian noise on range.

5. Geological transfer without training labels

Vaches Noires landslide, Normandy. Two visible UAV patches, 512×512 pixels, supplied by Cerema.



Test1: 47% / 50%



Test2: 29% / 34%

Legend: — Ours — U-Net + transfer learning — Ground truth

Jaccard:
ours / U-Net.
U-Net adapted on a
separate nearby image.

U-Net adaptation required about **two days of expert annotation**. Our method uses no training annotation. Two patches and one wall example are initial transfer tests, not a general validation.

References

- [1] Frangi et al., *Multiscale vessel enhancement filtering*, MICCAI, 1998. [2] Zhou et al., *FIND*, Zenodo, 2022.
[3] Jiang et al., *CrackSegDiff*, arXiv:2410.08100, 2024. [4] Ge et al., *Fine-tuning vision foundation model for crack segmentation in civil infrastructures*, Constr. Build. Mater. 431, 136573, 2024.

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Code & data